ELEN 635 10/15/2025

Review for Exam #1

All Students: Only the exam, pencil, unmarked paper and one calculator are allowed. No sharing a calculator is allowed. The equation sheet attached to this review will be provided on the test.

Distance Ed Instructions: If you have an on campus address you are required to take the exam in class.The exam will be held Wednesday evening at 8:00 PM.The exam will be posted on Canvas, and open 8 minutes early. You then print out the exam. After 50 minutes the test will shut off on Canvas and you will have 10 minutes take it to the printer and upload it to the Canvas Test module. The procedure is the same as it is with homework. Failure to upload in 10 minutes will incur a serious points penalty. Your cell phone should be on at the website used for office hours the entire time you are taking the exam. It should be propped up horizontally so that you and the screen are visible. Take the cell phone with you to the printer. I will come on to the Zoom site early so you can set up.

1. Know Maxwell's equations and the continuity condition in the frequency domain and how to derive the divergence equations using the curl equations and continuity condition. Know the names and units of each term in the equations.
2. Problems containing vector calculations: cross and dot products, curl, gradient and divergence.
3. Starting with Maxwell's equations containing electric current, derive the homogeneous vector Helmholtz equation.
4. For the homogeneous Helmholtz' equation, give the plane wave solution to this equation in its most general form and identify the terms.
5. Show that the expression for a plane wave satisfies Helmholtz equation.
6. Given a field in the frequency domain be able to convert it to the time domain and vice versa.
7. Given a general expression for the electric field find the magnetic field and vice versa.
8. Given an expression for a field identify its wave vector, position vector, wave number, direction of travel, velocity frequency, amplitude, and wavelength.
9. Given the electric field in the frequency domain show mathematically that the field given is a straight line, a circle or an ellipse, find an expression for the angle ψ between the field vector and the axis and describe the type of polarization (linear, circular, elliptical).
10. Given a field in a lossy media find the time average power density, be able to calculate the phase velocity, wavelength , loss tangent and skin depth.
11. Show that $\mathbf{E} = \mathbf{E}_o e^{-j\mathbf{k}\cdot\mathbf{r}}$ satisfies the vector Helmholtz equation $\nabla^2\mathbf{E} + k^2\mathbf{E} = 0$.
12. Given a field normally incident, perpendicularly or parallel polarized be able to find the reflected or transmitted field, Brewster and angle and critical angle.
13. Given the angle of incidence and amplitude of a parallel or perpendicularly polarized field find the reflected or transmitted field.
14. Be able to solve problems involving normal incidence on a layered medium using the transmission line method.

The following equation pages will be attached to the exam.

Useful equations

Cartesian Coordinates (x, y, z)

$$\nabla V = \mathbf{a}_x \frac{\partial V}{\partial x} + \mathbf{a}_y \frac{\partial V}{\partial y} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \mathbf{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \mathbf{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \mathbf{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Cylindrical Coordinates (r, ϕ, z)

$$\nabla V = \mathbf{a}_r \frac{\partial V}{\partial r} + \mathbf{a}_\phi \frac{\partial V}{r \partial \phi} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} \left(r A_r \right) + \frac{\partial A_\phi}{r \partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \mathbf{a}_r \left(\frac{\partial A_z}{r \partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \mathbf{a}_\phi \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) + \mathbf{a}_z \frac{1}{r} \left[\frac{\partial}{\partial r} \left(r A_\phi \right) - \frac{\partial A_r}{\partial \phi} \right]$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical Coordinates (R, θ, ϕ)

$$\nabla V = \mathbf{a}_R \frac{\partial V}{\partial R} + \mathbf{a}_\theta \frac{\partial V}{R \partial \theta} + \mathbf{a}_\phi \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 A_R \right) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} \left(A_\theta \sin \theta \right) + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \mathbf{A} = \mathbf{a}_R \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} \left(A_\phi \sin \theta \right) - \frac{\partial A_\theta}{\partial \phi} \right]$$

$$+ \mathbf{a}_\theta \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} \left(R A_\phi \right) \right]$$

$$+ \mathbf{a}_\phi \frac{1}{R} \left[\frac{\partial}{\partial R} \left(R A_\theta \right) - \frac{\partial A_R}{\partial \theta} \right]$$

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

(1) $A^2 = \mathbf{A} \cdot \mathbf{A}$

(2) $|A|^2 = \mathbf{A} \cdot \mathbf{A}^*$

(3) $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$

(4) $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$

(5) $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$

(6) $(\mathbf{A} + \mathbf{B}) \cdot \mathbf{C} = \mathbf{A} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{C}$

(7) $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \mathbf{B} \cdot \mathbf{C} \times \mathbf{A} = \mathbf{C} \cdot \mathbf{A} \times \mathbf{B}$

(9) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$

(10) $\nabla(v + w) = \nabla v + \nabla w$

(11) $\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}$

(12) $\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$

(13) $\nabla(vw) = w \nabla v + v \nabla w$

(14) $\nabla \cdot (w\mathbf{A}) = w \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla w$

(15) $\nabla \times (w\mathbf{A}) = w \nabla \times \mathbf{A} - \mathbf{A} \times \nabla w$

(16) $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B}$

(17) $\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times \nabla \times \mathbf{A}$

(18) $\nabla \times (v \nabla w) = \nabla v \times \nabla w$

(19) $\nabla \times \nabla w = 0$

(20) $\nabla \cdot \nabla \times \mathbf{A} = 0$

$$x = \rho \cos \phi$$
$$y = \rho \sin \phi$$
$$z = z$$

$$x = r \sin \theta \cos \phi$$
$$y = r \sin \theta \sin \phi$$
$$z = r \cos \theta$$

$\int_v \nabla \cdot \mathbf{A} dv = \oint_s \mathbf{A} \cdot ds$ (Divergence theorem)

$\int_s \nabla \times \mathbf{A} \cdot ds = \oint_c \mathbf{A} \cdot d\ell$ (Stokes's theorem)

Useful Equations

$$\Gamma_\perp = \frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t} \quad ; \quad \tau_\perp = \frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$$

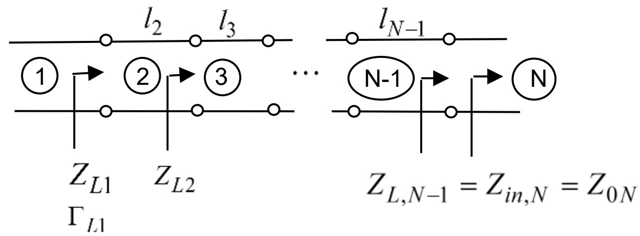
$$\Gamma_\parallel = \frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i} \quad ; \quad \tau_\parallel = \frac{2\eta_2 \cos\theta_i}{\eta_1 \cos\theta_i + \eta_2 \cos\theta_t}$$

$$\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$$

$$\hat{\mathbf{n}} \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$$

$$\hat{\mathbf{n}} \cdot (\varepsilon \mathbf{E}_1 - \varepsilon \mathbf{E}_2) = \rho_s$$

$$\hat{\mathbf{n}} \cdot (\mu \mathbf{H}_1 - \mu \mathbf{H}_2) = 0$$



$$\alpha = \omega \left[\frac{\mu \varepsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \varepsilon} \right)^2} - 1 \right) \right]^{1/2}$$

$$\beta = \omega \left[\frac{\mu \varepsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \varepsilon} \right)^2} + 1 \right) \right]^{1/2}$$

$$\eta_c = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \frac{j\omega\mu}{\alpha + j\beta}$$

$$\tan\delta = \frac{\sigma}{\omega\varepsilon}$$

$$V_1 = V_{01} e^{-jk_1 z} \left(1 + \Gamma_{L1} e^{j2k_1 z} \right)$$

$$V_n = V_{0n} e^{-jk_n (z - d_n)} \left(1 + \Gamma_{Ln} e^{j2k_n (z - d_n)} \right)$$

$$d_n = \sum_{m=1}^{n} l_m \quad \text{with } l_1 = 0$$

$$V_N = V_{0N} e^{-jk_N (z - d_{N-1})}$$

$$\Gamma_{Ln} = \frac{Z_{Ln} - Z_{on}}{Z_{Ln} + Z_{on}}$$

$$Z_{L,n} = Z_{o,n+1} \frac{Z_{L,n+1} + jZ_{o,n+1} \tan(k_{n+1} l_{n+1})}{Z_{o,n+1} + jZ_{L,n+1} \tan(k_{n+1} l_{n+1})}$$

$$Z_{L,N-1} = Z_{o,N}$$